CLINICAL · NECK, THYROID & SALIVARY

Head and neck reconstruction and facial reanimation

A head and neck resection is only half the operation; the other half is putting back what was taken, and the patient lives with the second half. Reconstruction starts with a defect analysis: which tissues are missing (skin, lining, muscle, bone, nerve), what each one did, and what the bed is like. Choose the simplest method that restores both function and appearance. Today that is often a free flap at the same operation. Facial reanimation follows the same logic: what is missing, and for how long?

AT A GLANCE

- Concave sites heal well by secondary intention.
- Abbe for a third to a half of the lip; Karapandzic up to two-thirds.
- Fibula for the mandible; radial forearm for the tongue; check the Allen test.
- Facial muscle survives about 18 to 24 months of denervation.

1 CHAPTER ONE Principles

Defect analysis and the ladder.

1.1 What is missing

Every defect is analysed in the same way. First, which tissues have gone: skin, mucosal lining, muscle, bone, cartilage or nerve. Second, what each did, because the reconstruction must restore function as well as form: a skeleton for occlusion, contour and *dental implants*; cover for vital structures such as the carotid, dura and brain; oral competence and a lip sphincter; pliable, mobile tissue for tongue movement, speech and swallowing; and volume, sensation and secreting mucosa. Third, the state of the bed: irradiation, infection, a fistula and the vessels available, remembering that irradiated or infected beds need *vascularised* tissue. Finally, the patient's health, prognosis and wishes.

1.2 The reconstructive ladder (and the elevator)

1. **Healing by secondary intention:** good for *concave* sites (medial canthus, temple, conchal bowl, nasolabial groove, glabella); bad near *free margins* (eyelid, ala, lip), which contract and distort.
2. **Primary closure:** along relaxed skin tension lines, as a fusiform excision about three times as long as it is wide.
3. **Grafts:** skin, cartilage, bone, nerve and composite grafts.
4. **Local flaps.**
5. **Regional (pedicled) flaps.**
6. **Free tissue transfer.**
7. **Prostheses.**

Climbing the ladder rung by rung is not compulsory. The surgeon may take the "elevator" straight to the option that gives the best result, which for a major defect is often a free flap.

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2 CHAPTER TWO Grafts

Tissue without its own blood supply.

2.1 Skin grafts

A skin graft survives for about 48 hours by plasmatic *imbibition*, and then by *inosculation* of its vessels with those of the bed and capillary ingrowth at three to five days. Anything that separates graft from bed in that time kills it: haematoma, seroma, *shearing* and infection, which are prevented with a bolster or tie-over dressing, quilting sutures, pie-crusting or negative pressure. Grafts do not take on bare bone or cartilage stripped of periosteum or perichondrium, nor on irradiated beds.

Split-thickness grafts, taken with a dermatome from the thigh or buttock, take more reliably but *contract more* and match colour poorly; they line the mouth and maxillectomy cavities and cover the scalp over periosteum and free flap donor sites. **Full-thickness grafts**, defatted and taken from postauricular, preauricular, upper

eyelid or supraclavicular skin, contract less and match better, so they are used on the nose, eyelids and ear.

2.2 Cartilage, bone and nerve grafts

Septal cartilage is harvested leaving an L-strut of 1 to 1.5 cm; curved **conchal cartilage** suits the ala and ear; and **costal cartilage** is plentiful for the dorsum and microtia but can warp. **Composite** skin–cartilage grafts from the ear repair alar rim and columella defects up to about 1 to 1.5 cm.

Cortical bone from split calvaria, rib or iliac crest rebuilds the orbital floor, nasal dorsum and facial buttresses, and **cancellous bone** from the iliac crest fills alveolar clefts and small mandibular gaps. Every free bone graft needs a well-vascularised, unirradiated bed and rigid fixation, and all resorb to some degree.

Nerve grafts come from the great auricular nerve, which lies in the operative field and gives up to about 10 cm; the sural nerve, up to 30 to 35 cm, used for cross-face grafts; and the medial antebrachial cutaneous nerve.

OPERATIVE GUIDES IN THEATRE 5 operations

Skin grafting, fascial and allografts, calvarial and rib grafts, and nerve grafting.

- Skin Grafting
- Autologous Fascial Grafts and Acellular Allografts
- Split Calvarial Graft
- Rib Graft
- Nerve Graft Techniques

3 CHAPTER THREE Local flaps

Like for like.

3.1 Principles

Random-pattern local flaps live on the subdermal plexus. On the well-vascularised face they tolerate a length-to-width ratio of up to about 3:1, more than on the trunk. They are designed to place scars in the relaxed skin tension lines and at the borders

of *aesthetic subunits*, to borrow skin from areas of laxity such as the cheek, glabella and neck, and to close the secondary defect without distorting a free margin. Smoking, tension, irradiation and haematoma are their enemies.

3.2 Types

Advancement flaps include the V-Y and the unipedicle and bipedicle flaps of the forehead and of the lip with a crescentic perialar excision. **Rotation** flaps include the cervicofacial and scalp flaps. **Transposition** flaps include the *rhomboid (Limberg)* flap, the *bilobed (Zitelli)* flap for defects up to about 1.5 cm on the lower third of the nose, the *nasolabial (melolabial)* flap for the ala, and the note flap.

Interpolated flaps pass over or under intact skin and are divided later. The *paramedian forehead flap*, on the *supratrochlear artery* and divided at three weeks, is the workhorse for large or full-thickness nasal defects; the interpolated nasolabial flap and the *Abbe* cross-lip flap are the others. A full-thickness nasal defect needs all *three layers* rebuilt: lining from local mucosal flaps or a turnover flap, a framework of cartilage grafts, and cover from the forehead flap.

OPERATIVE GUIDES IN THEATRE 4 operations

Paramedian forehead, cervicofacial and scalp flaps, and nasal reconstruction.

Paramedian Forehead Flaps
Forehead flap
Cervicofacial Transposition Flaps
Scalp Reconstruction: Role of Tissue Expansion and Flap Reconstruction
Nasal Reconstruction

3.3 Lip reconstruction

FULL-THICKNESS LIP DEFECTS

DEFECT	RECONSTRUCTION
Under a third (up to about a half of the lower lip)	Primary (V or W) closure
A third to a half	Abbe cross-lip flap (on the labial artery, divided at about three weeks); Estlander flap if the commissure is involved

DEFECT	RECONSTRUCTION
A half to two-thirds	Karapandzic (neurovascular, sensate, innervated rotation–advancement); bilateral advancement
Over two-thirds to total	Bernard–Burow (cheek advancement), Gillies fan flap, or a free radial forearm flap with a palmaris longus sling

What matters most is restoring the *orbicularis sphincter*, the vermilion line and the commissure. *Microstomia* is the price paid for large local flaps.

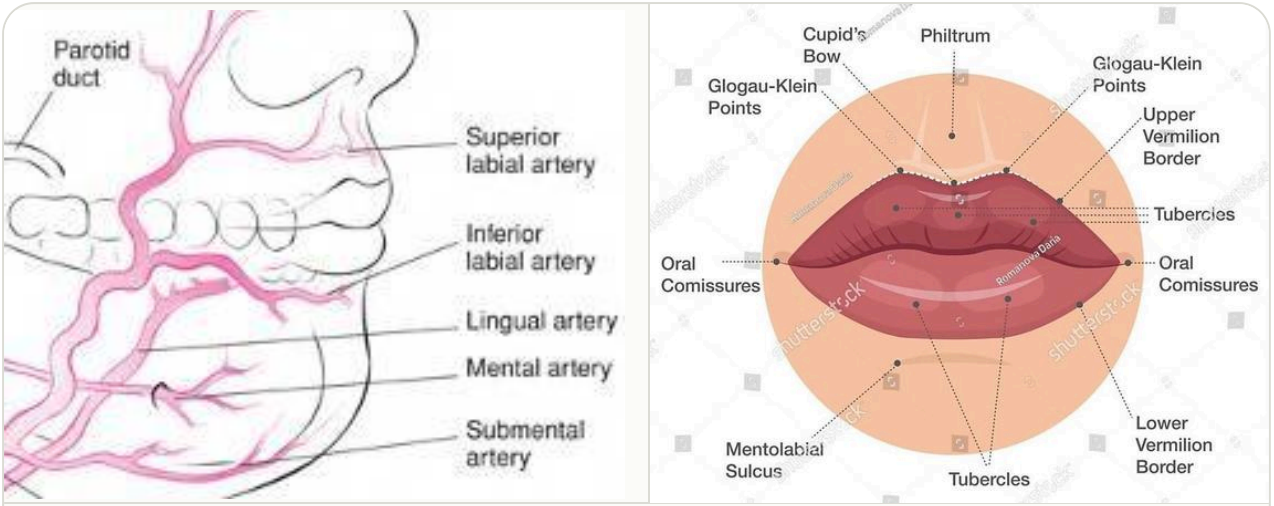


FIGURE 1 The arterial supply of the lips, and the lip landmarks.
Anatomy atlas · ENT Steps p. 168, ENT Steps p. 166

4 CHAPTER FOUR Regional flaps

Pedicled from nearby.

REGIONAL FLAPS

FLAP	PEDICLE	USES AND NOTES
Pectoralis major myocutaneous	Pectoral branch of the *thoracoacromial artery*	Workhorse salvage flap: oral, pharyngeal and neck defects, carotid cover; reaches about to the zygomatic arch; bulky, can sag
Deltopectoral (fasciocutaneous)	Perforators of the internal thoracic (mammary) artery, 2nd to 4th intercostal spaces	Neck skin, pharyngostome; often needs delay and a second stage

FLAP	PEDICLE	USES AND NOTES
Temporalis muscle	Deep temporal arteries	Orbit, maxillectomy, skull base, facial reanimation
Temporoparietal fascia	Superficial temporal artery	Ear, orbit and skull base; thin and vascular; can be free
Supraclavicular island	Supraclavicular branch of the transverse cervical artery	Thin, pliable neck and face skin; pharynx
Submental island	Submental branch of the facial artery	Oral cavity and face (beware nodes in oral cancer)
Sternocleidomastoid	Occipital, superior thyroid, transverse cervical	Small buttress; parotid contour
Trapezius (lateral, lower island)	Transverse cervical and dorsal scapular arteries	Posterior neck, scalp
Latissimus dorsi (pedicled)	Thoracodorsal artery	Large defects of neck and scalp
Nasoseptal, turbinate, pericranial	Septal artery, sphenopalatine branches, supraorbital and supratrochlear	Skull base
Palatal island	Greater palatine artery	Palate and nasopharynx

OPERATIVE GUIDES IN THEATRE 5 operations

Pectoralis major, deltopectoral, trapezius and latissimus, temporoparietal fascia and temporalis flaps.

Pectoralis Major Musculocutaneous Flaps in Head and Neck Surgery
Pectoralis major
Deltopectoral Flap
Trapezius and Latissimus Dorsi Regional Flaps
Temporoparietal Fascial Flap
Temporalis Muscle Transposition

5 CHAPTER FIVE Free flaps

Microvascular transfer.

5.1 Principles

Free flaps are joined to recipient arteries such as the facial, superior thyroid, lingual or transverse cervical, and to the internal jugular vein end-to-side or the external jugular or facial veins; a previously treated neck may need a contralateral approach or vein grafts. Two teams work together, ischaemia time is kept short (bowel is the most sensitive, then muscle, then bone), and the pedicle is laid out without kinking, twisting or tension. Success rates are about 95 to 99%.

Most failures occur within 72 hours, usually from venous thrombosis, so the flap is checked hourly at first for colour, capillary refill, temperature and pinprick bleeding, with a hand-held or implantable Doppler. The patient is kept warm, well filled and normotensive, vasoconstrictors are avoided where possible, and aspirin or heparin is given according to unit policy. **Venous congestion** makes the flap purple and swollen with brisk dark bleeding; **arterial insufficiency** makes it pale and cool with no bleeding. Either means an urgent return to theatre, because salvage is far more likely within hours.

5.2 The workhorse flaps

COMMON FREE FLAPS

FLAP	PEDICLE	TISSUE	TYPICAL USE
Radial forearm	Radial artery and venae comitantes, cephalic vein	Thin, pliable fasciocutaneous; sensate; can carry radius	Tongue, floor of mouth, palate, pharynx (tubed), lip; check the *Allen test*
Anterolateral thigh	Descending branch of the lateral circumflex femoral artery (perforators)	Skin, fat, fascia, vastus lateralis; adjustable bulk	Tongue (total glossectomy), cheek, skull base, pharynx; hidden donor site; variable perforators
Fibula	Peroneal artery	Up to about 25 cm of straight bone, segmented by osteotomy; skin paddle	*Mandible*, maxilla; implant-capable (double-barrel for height); image leg vessels first (peronea arteria magna)

FLAP	PEDICLE	TISSUE	TYPICAL USE
Scapula and parascapular (subscapular system)	Circumflex scapular, thoracodorsal (angular branch to tip)	Bone, skin, latissimus, serratus in combinations ("chimeric")	Complex 3D maxillary and mandibular defects; elderly with poor leg vessels
Iliac crest	Deep circumflex iliac artery	Thick curved bone with internal oblique	Mandible, maxilla; donor pain and hernia
Rectus abdominis	Deep inferior epigastric artery	Muscle and skin; bulky	Skull base, maxillectomy, orbit, total glossectomy
Latissimus dorsi	Thoracodorsal artery	Large thin muscle	Scalp, large surface defects
Jejunum	Mesenteric arcade	Tubed, secretory mucosa	Circumferential pharyngo-oesophageal defects; peristalsis and voice quality are drawbacks
Gracilis	Medial circumflex femoral artery; anterior branch of the obturator nerve	Functioning muscle	*Facial reanimation*

5.3 Choosing the flap

In the **mandible**, a lateral segmental defect is usually rebuilt with a fibula, though a plate with a soft-tissue flap may suit a patient with a poor prognosis. An anterior arch defect must always be rebuilt with bone, or the "*Andy Gump*" deformity follows: a collapsed chin, drooling and loss of oral competence. The **maxilla** is restored with an obturator prosthesis or a free flap (fibula, scapula or anterolateral thigh), chosen by the size of the defect and the involvement of the palate and orbital floor.

The **tongue** needs thin, mobile tissue after partial glossectomy, so a radial forearm flap is used, but bulk after total glossectomy, so an anterolateral thigh or rectus flap fills the mouth and reaches the palate. In the **pharynx**, partial defects take a patch of radial forearm or anterolateral thigh, while circumferential defects need a tubed anterolateral thigh or radial forearm flap, jejunum, or a gastric pull-up when the oesophagus is removed.

OPERATIVE GUIDES IN THEATRE 3 operations

Microvascular reconstruction, and marginal and segmental mandibulectomy.

Microvascular Reconstruction of the Head and Neck
Free flaps
Marginal Mandibulectomy
Segmental Mandibulectomy

6 CHAPTER SIX Facial reanimation

Time decides the method.

6.1 Questions to ask

1. What caused the palsy, and is the nerve in continuity?
2. Is the proximal stump available?
3. *How long has the face been paralysed?* Facial muscles survive denervation for about 18 to 24 months, after which motor end plates are lost and muscle atrophies.
4. Which regions are affected, and what does the patient want: eye protection, symmetry at rest, a smile?

6.2 Protect the eye first

Whatever else is planned, the cornea is protected first: lubricating drops by day and ointment at night, taping and a moisture chamber. A *gold or platinum weight* in the upper lid helps it close, a lateral tarsal strip or canthoplasty tightens a lax lower lid, a brow lift corrects brow ptosis, and a tarsorrhaphy is done if the cornea is at risk. Botulinum toxin to the levator produces a temporary protective ptosis.

6.3 Restoring the nerve (within about 18 to 24 months)

A nerve cut at the time of injury is repaired at once by tension-free epineurial **neurorrhaphy**, and any transection in the temporal bone or at parotidectomy is repaired immediately. Where the gap is 1 cm or more, an **interposition graft** of great auricular or sural nerve is used; function returns at 6 to 12 months, usually with

some *synkinesis*. Branches medial to a vertical line through the lateral canthus are usually not repaired, because of cross-innervation.

When the proximal stump is unavailable but the distal nerve and muscle are still viable, a **nerve transfer** is used. The *masseteric-to-facial* (V to VII) transfer is now the commonest: fast, strong and with little morbidity, though the smile is driven by biting and is usually not spontaneous. The *hypoglossal-to-facial* (XII to VII) transfer, done as a partial split or jump-graft technique to limit tongue atrophy, gives good tone but synkinesis. *Cross-face nerve grafting* carries axons from the contralateral buccal branches through a sural graft, with a second stage once Tinel's sign shows they have crossed at about 8 to 12 months. It is the only method that gives a *spontaneous, emotional smile*, but it is weaker, so it is often combined with a masseteric transfer acting as a "*babysitter*" to keep the muscles alive.

6.4 Long-standing palsy (muscle lost)

Once the facial muscles are lost, a new muscle must be brought in. The **free functional gracilis transfer**, powered by the masseteric nerve in one stage or by a cross-face graft in two, is the gold standard for smile reanimation and is also used in *Möbius syndrome*. The **temporalis transfer**, as an orthodromic lengthening temporalis myoplasty from the coronoid or as the older turnover over the arch, is the alternative. **Static procedures** add support: fascia lata or acellular dermal slings to lift the commissure and support the nasal valve, a brow lift, and lower lid tightening.

6.5 Synkinesis and hyperkinesis

Aberrant regeneration after Bell's palsy or nerve repair produces involuntary co-contractions, such as the eye closing on smiling, and hypertonicity. These are treated with *neuromuscular retraining* by facial physiotherapy, *botulinum toxin* to the overactive muscles on the affected side and to the normal side for symmetry, and selective neurectomy or myectomy.

OPERATIVE GUIDES IN THEATRE 2 operations

Facial reanimation, and eyelid reanimation.

Facial Reanimation
Facial reanimation

Eyelid Reanimation in Patients With Facial Paralysis
Eyelid reanimation

7 CHAPTER SEVEN Other options

Prostheses and beyond.

7.1 Beyond tissue transfer

Auricular, nasal and orbital **prostheses** retained by *osseointegrated implants* or adhesives, and palatal *obturators* after maxillectomy, suit the elderly and the irradiated, follow failed reconstruction, and allow the cavity to be inspected for recurrence. **Tissue expansion** recruits skin for the scalp and neck. **Virtual surgical planning** uses CT-based 3D models, cutting guides and custom plates for mandibular and maxillary reconstruction. **Facial allotransplantation** of composite tissue has been done for devastating defects, but lifelong immunosuppression, rejection and ethical questions still limit it.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The reconstructive ladder;** sites that heal well by secondary intention.
- **Split- versus full-thickness grafts:** take, contraction, uses.
- **Bilobed, rhomboid and paramedian forehead flaps:** blood supply, three-layer nasal repair.
- **Lip reconstruction by size of defect:** Abbe, Estlander, Karapandzic.
- **The pectoralis major flap:** its pedicle and reach.
- **Radial forearm, anterolateral thigh and fibula flaps:** pedicles, uses, and pre-operative checks (Allen test, leg vessel imaging).

- **Free flap monitoring** and the failing flap.
- **Facial reanimation by time since injury;** masseteric versus hypoglossal versus cross-face grafting; the gracilis flap; the eye first.

8.2 Common pitfalls

COMMON PITFALLS

- Skin grafting onto irradiated tissue or bare bone.
- Leaving an anterior mandibular defect unreconstructed.
- Raising a radial forearm flap without an Allen test.
- Delaying return to theatre for a congested free flap.
- Waiting beyond two years before trying to reinnervate the face.

Written for this app from Oxford Textbook of Otolaryngology ch. 64, Oxford Specialist Handbook ch. 9, K. J. Lee ch. 55. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

ORAL CAVITY & TONGUE · AFTER SURGERY

Squamous cell carcinoma: oral cavity, oropharynx and larynx

IMAGING FINDINGS

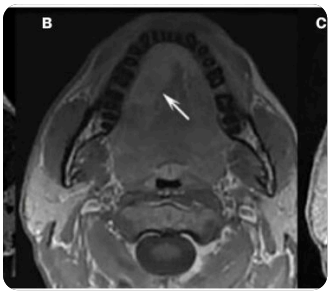
- Enhancing mucosal mass; depth of invasion in the tongue
- Crossing the lingual septum; neurovascular bundles
- Mandible cortex (CT) and marrow (MRI)

TOP DIFFERENTIALS

- Lingual tonsil hyperplasia
- Abscess
- Minor salivary gland tumour

CHECKLIST

- MRI for oral and oropharyngeal tumours
- Nodes on both sides; chest CT



Axial MRI of an oral cavity SCC shows a solid mass involving the left lateral border of the tongue showing intermediate signal intensity on T2…



Coronal T1WI MR in a woman with a painful right tongue ulcer shows subtle low signal intensity at the lateral tongue margin. There is no evidence …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504.